

Policy Gradient RL

$$s, a, \pi(a/s), r(s, a), p(s'/s, a)$$

$$\tau/\pi = \{s_0, a_0, r_0, s_1, a_1, r_1, \dots\}$$

$$\cdot p(\tau/\pi) = p(s_0) \pi(a_0/s_0) p(s_1/s_0, a_0) \pi(a_1/s_1) \dots$$

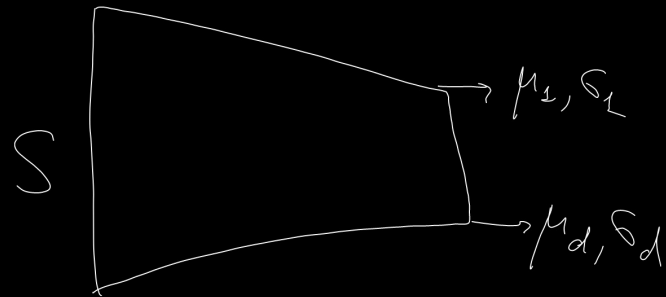
$$\cdot R_0 = r_0 + \gamma r_1 + \gamma^2 r_2 + \dots, \quad \gamma \in (0, 1)$$

$$J(\pi) = \mathbb{E}_{\tau/\pi} R_0(\tau) \rightarrow \max_{\pi}$$

In DQN: $\arg\max_a Q(s, a)$

$$\pi(a|s, \theta) = \prod_{j=1}^d \mathcal{N}(a_j | \mu_j(s, \theta), \log(1 + \exp(\sigma_j(s, \theta))))$$

$a \in \mathbb{R}^d$



$$a \in \{1, 2, \dots, K\}$$

$$\Rightarrow \pi(a|s, \theta) = \text{softmax}(\text{outputs}(s, \theta))$$

$$J(\theta) = E_{p(\tau|\theta)} R(\tau) \rightarrow \max_{\theta}$$

$$\nabla_{\theta} J(\theta) = \nabla_{\theta} E_{p(\tau|\theta)} R(\tau) = \left\{ \begin{array}{l} \text{log deriv.} \\ \text{trick} \end{array} \right\} =$$

$$= \nabla_{\theta} \int p(\tau|\theta) R(\tau) d\tau = \int \nabla_{\theta} p(\tau|\theta) R(\tau) d\tau =$$

$$= \int p(\tau|\theta) \left(\frac{\nabla_{\theta} p(\tau|\theta)}{p(\tau|\theta)} \right) R(\tau) d\tau = E_{p(\tau|\theta)} \nabla_{\theta} \log p(\tau|\theta) R(\tau)$$

$\frac{\nabla_{\theta} p(\tau|\theta)}{p(\tau|\theta)} = \nabla_{\theta} \log p(\tau|\theta)$

$$\tau := \{s_0, a_0, s_1, a_1, \dots, s_T, a_T\}$$

$$\begin{aligned} \mathbb{E}_{p(\tau|\theta)} \nabla_{\theta} \log p(\tau|\theta) \underbrace{R(\tau)} &= \left\{ p(\tau|\theta) = p(s_0) \pi(a_0|s_0, \theta) \times \right. \\ &\quad \left. \times p(s_1|s_0, a_0) \pi(a_1|s_1, \theta) \dots \right\} = \\ &= \sum_{t=0}^T \mathbb{E}_{p(\tau|\theta)} \nabla_{\theta} \log \pi(a_t|s_t, \theta) \left(\sum_{u=0}^T \gamma^u r_u \right) = \\ &= \sum_{t, u=0}^T \mathbb{E}_{p(\tau|\theta)} \nabla_{\theta} \log \pi(a_t|s_t, \theta) \gamma^u r_u \end{aligned}$$

$$\underline{u < t} \quad \mathbb{E}_{p(\tau|\theta)} \nabla \log \pi(a_t | s_t, \theta) \gamma^u z_u =$$

$$= \mathbb{E}_{s_0, a_0, s_1, a_1, \dots, s_u, a_u} \gamma^u z_u \mathbb{E}_{s_{u+1}, a_{u+1}, \dots} \left(\mathbb{E}_{a_t \sim \pi(a_t | s_t, \theta)} \nabla \log \pi(a_t | s_t, \theta) \right)$$

$$\begin{aligned} \mathbb{E}_{a_t \sim \pi(a_t | s_t, \theta)} \nabla_{\theta} \log \pi(a_t | s_t, \theta) &= \\ \sum_{a_t} \cancel{\pi(a_t | s_t, \theta)} \frac{(\nabla_{\theta} \pi(a_t | s_t, \theta))}{\cancel{\pi(a_t | s_t, \theta)}} &= \nabla_{\theta} \sum_{a_t} \pi(a_t | s_t, \theta) = \nabla_{\theta} 1 = 0 \end{aligned}$$

$$\nabla_{\theta} J(\theta) = \sum_{t=0}^T E_{p(\tau|\theta)} \nabla_{\theta} \log \pi(a_t | s_t, \theta) \gamma \left(\sum_{u=t}^T \gamma^u z_u \right)$$

weighted ML maximization

REINFORCE algorithm

Init. θ

For episodes:

Init. s_0

For $t=0, 1, 2, \dots, T$:

$a_t \sim \pi(a | s_t, \theta)$, get r_t, s_{t+1}

$R=0, L_{\theta}=0$

For $t=T, T-1, \dots, 0$:

$R = r_t + \gamma R$

$L = L - \log \pi(a_t | s_t, \theta) \gamma^t R$

$\theta = \text{opt_step}(\nabla_{\theta} L)$

Limitations: use Q instead of R

① Full episode

② Huge variance of SG
→ use baseline

← can be omitted in practice

Actor-Critic Approach

$Q(s, a | \theta_Q)$ - critic

$\pi(a | s, \theta_\pi)$ - actor

Th (PG theorem)

$$J(\theta) = \mathbb{E}_{p(s_0)} \mathbb{E}_{a_0 \sim \pi(a_0 | s_0, \theta)} Q^{\pi_\theta}(s_0, a_0)$$

$$\Rightarrow \nabla_{\theta} J(\theta) = \frac{1}{1-\gamma} \mathbb{E}_{p(s_0)} \mathbb{E}_{s \sim \mu^{\pi}(s)} \mathbb{E}_{a \sim \pi(a | s, \theta)} \nabla_{\theta} \log \pi(a | s, \theta) Q^{\pi_\theta}(s, a)$$

state visitation
distribution

$$\mu^{\pi}(s) = \sum_{t=0}^{\infty} \gamma^t p(s_0 \rightarrow s | t, \pi)$$

$$\mathbb{E}_{s,a} \nabla \log \pi(a|s, \theta) (Q(s,a) - \overset{\text{baseline}}{B(s)})$$

$$\mathbb{E}_s \mathbb{E}_{a \sim \pi(a|s)} \nabla \log \pi(a|s, \theta) (B(s)) = 0$$

Good choice for baseline: $B(s) = \mathbb{E}_{a \sim \pi(a|s)} Q^\pi(s,a) = V^\pi(s)$

Example $E_a f_1(a) \cdot f_2(a)$

$$a: \begin{array}{cc} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{array} \quad f_1 = \begin{cases} 1, & \text{if } a=0 \\ -1, & \text{if } a=1 \end{cases} \quad f_2 = \begin{cases} 100, & \text{if } a=0 \\ 101, & \text{if } a=1 \end{cases}$$

$$E_a f_1 = \frac{1}{2} \cdot 1 + \frac{1}{2}(-1) = 0$$

$$E_a f_1 f_2 = \frac{1}{2} (1 \cdot 100) + \frac{1}{2} (-1) \cdot 101 = -\frac{1}{2}$$

$$\text{Var} = E_a (f_1 f_2 - E f_1 f_2)^2 = \frac{1}{2} \left(1 \cdot 100 + \frac{1}{2} \right)^2 + \frac{1}{2} \left((-1) \cdot 101 + \frac{1}{2} \right)^2 = 100.5^2$$

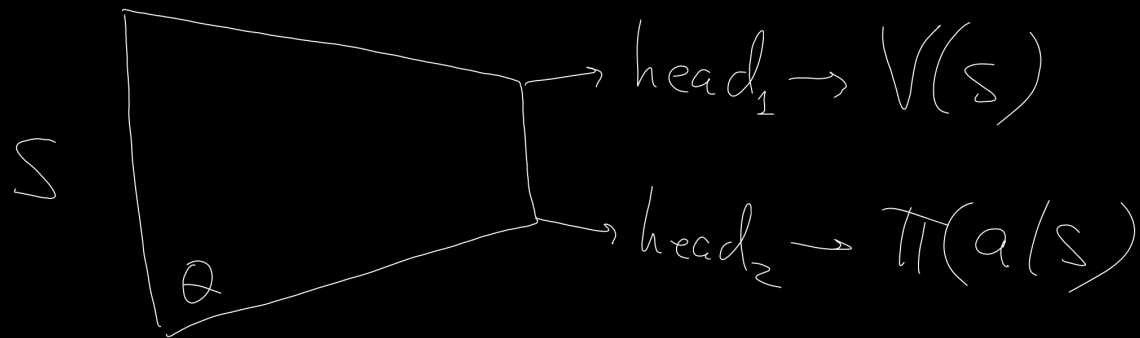
$$E_a p_1 (f_2 - E f_2)$$

$$E f_2 = \frac{1}{2} 100 + \frac{1}{2} 101 = 100.5$$

$$\begin{aligned} \text{Var} &= E_a \left(p_1 (f_2 - E f_2) - (E f_1 - f_2) \right)^2 = \\ &= \frac{1}{2} \left(1 (100 - 100.5) + \frac{1}{2} \right)^2 + \frac{1}{2} \left((-1)(101 - 100.5) + \frac{1}{2} \right)^2 = 0 \end{aligned}$$

$$A^\pi(s, a) := Q^\pi(s, a) - V^\pi(s) \text{ - advantage}$$

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{p(s'|s, a)} V^\pi(s')$$



A2C (Advantage Actor Critic)

Init. θ

For episodes:

Init. $s_0, k=0$

while $k' < T$:

for $t = k, k+1, \dots, k+n$:

$a_t \sim \pi(a_t | s_t, \theta)$, get z_t, s_{t+1}

$R = V(s_{k+n+1} | \theta), L = 0$

for $t = k+n, k+n-1, \dots, k$:

$R = z_t + \gamma R$

$L = L + \frac{1}{2} (V(s_t | \theta) - R)^2$

$L = L - \log \pi(a_t | s_t, \theta) (R - V(s_t, \theta)) - \lambda \text{Entropy}(\pi(\cdot | s_t, \theta))$

$\theta = \text{opt_step}(\nabla_{\theta} L)$

$k = k+n$

rollout size

use sync. parallel agents

detach

Off-policy vs. on-policy RL

$\pi(a|s, \theta)$ - learned policy

$b(s)$ - behaviour policy (Reinforce, A2C)

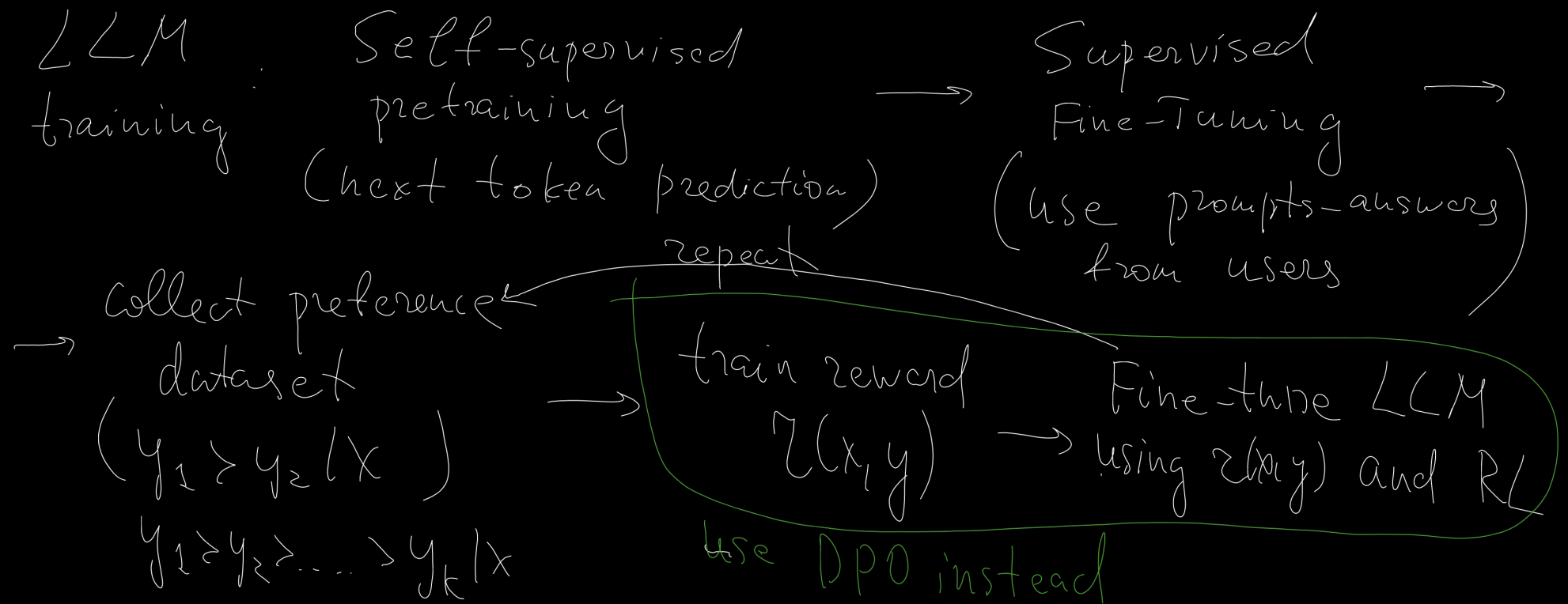
If $\pi(a|s, \theta) = b(s) \Rightarrow$ on-policy RL

Otherwise $\Rightarrow$ off-policy RL (DQN)

PPO - on-policy PG

SAC - off-policy PG

RLHF (RL from Human Feedback)



$$y_1 > y_2 / x$$

$$\text{Bradley-Terry model: } p(y_1 > y_2 / x) =$$

$$= \frac{\exp(z(x, y_1))}{\exp(z(x, y_1)) + \exp(z(x, y_2))} = \frac{1}{1 + \exp(z(x, y_2) - z(x, y_1))}$$

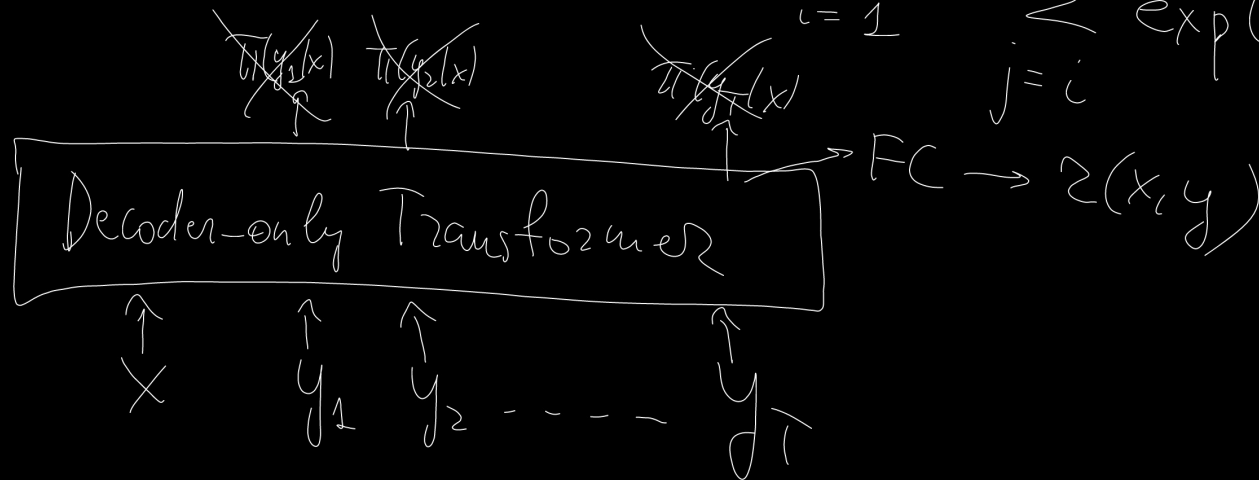
$$= \sigma(z(x, y_1) - z(x, y_2))$$

$$\text{Loss}(\varphi) = -\mathbb{E}_{x, y_1, y_2} \log \sigma(z_\varphi(x, y_1) - z_\varphi(x, y_2)) \rightarrow \min_{\varphi}$$

$$y_1 \succ y_2 \succ \dots \succ y_k \mid x$$

Plackett-Luce model:

$$p(y_1 \succ \dots \succ y_k \mid x) = \prod_{i=1}^k \frac{\exp(z(x, y_i))}{\sum_{j=i}^k \exp(z(x, y_j))}$$



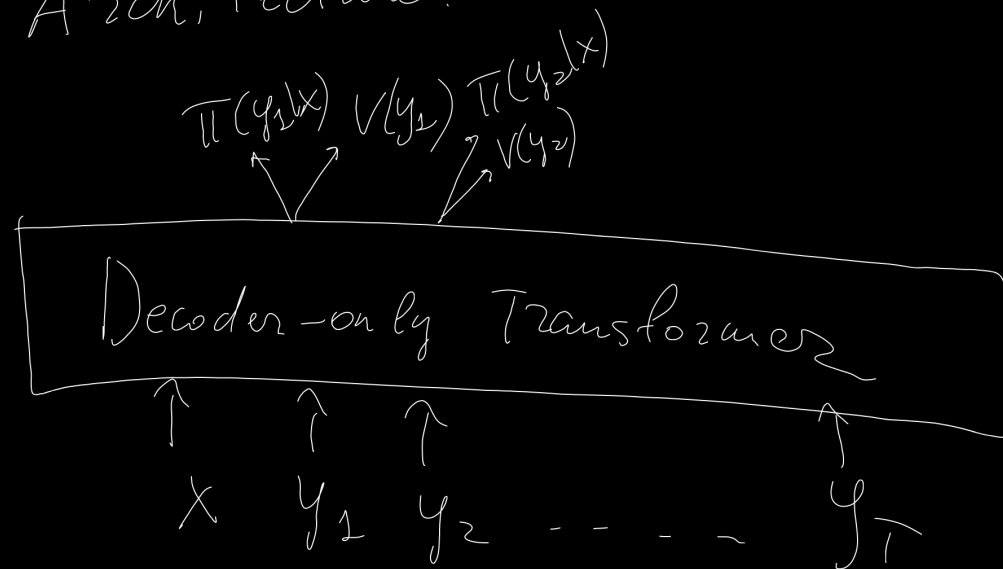
$$RL: \mathbb{E}_x \mathbb{E}_{y \sim \pi_\theta(y|x)} \left(r_\varphi(x, y) - \beta KL(\pi(y|x, \theta) \parallel \pi_{SFT}(y|x)) \right)$$

Episode in RL: $\rightarrow \max_{\theta}$

$$\begin{aligned} s_0 &= x, a_0 = y_1, \quad r(s_0, a_0) = -\beta KL(\pi(y_1|x, \theta) \parallel \pi_{SFT}(y_1|x)) \\ s_1 &= (x, y_1), a_1 = y_2, \quad r(s_1, a_1) = -\beta KL(\pi(y_2|y_1, x, \theta) \parallel \pi_{SFT}(y_2|x)) \\ &\dots \end{aligned}$$

$$s_T = (x, y_1, \dots, y_{T-1}), a_T = EOS, \quad r(s_T, a_T) = r_\varphi(x, y) - \beta KL(\pi(y_T|x, \theta) \parallel \pi_{SFT}(y_T|x))$$

Architecture:



Direct Preference Opt. (DPO)

$$\text{Fix } x: \mathbb{E}_{y \sim \pi(y|x)} \left(r(x, y) - \beta \text{KL}(\pi(y|x) \parallel \pi_{\text{ref}}(y|x)) \right)$$

$$\begin{cases} \sum_y \pi(y|x) (r(x, y) - \beta (\log \pi(y|x) - \log \pi_{\text{ref}}(y|x))) \xrightarrow{\max_{\pi}} \\ \sum_y \pi(y|x) = 1, \pi(y|x) \geq 0 \forall y \end{cases} \xrightarrow{\max_{\pi}}$$

$$\mathcal{L}(\pi, \mu) = \sum_y \pi(y|x) \left(z(x, y) - \beta (\log \pi(y|x) - \log \pi_{\text{ref}}(y|x)) \right) + \mu \left(\sum_y \pi(y|x) - 1 \right)$$

$$\frac{\partial \mathcal{L}(\pi, \mu)}{\partial \pi(y|x)} = z(x, y) - \beta (\log \pi(y|x) + 1 - \log \pi_{\text{ref}}(y|x)) + \mu = 0$$

$$\Rightarrow \log \pi(y|x) = \frac{1}{\beta} z(x, y) + \log \pi_{\text{ref}}(y|x) - 1 + \frac{\mu}{\beta}$$

$$\pi(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} z(x, y)\right)$$

$$Z(x) = \sum_y \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} z(x, y)\right)$$

$$\pi(y|x) = \frac{1}{z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} z(x, y)\right)$$

$$\log \pi(y|x) = -\log z(x) + \log \pi_{\text{ref}}(y|x) + \frac{1}{\beta} z(x, y)$$

$$z(x, y) = \beta (\log \pi(y|x) - \log \pi_{\text{ref}}(y|x)) + \beta \log z(x)$$

$$\sigma(z(x, y_1) - z(x, y_2)) = \sigma\left(\beta \log \frac{\pi(y_1|x)}{\pi_{\text{ref}}(y_1|x)} - \beta \log \frac{\pi(y_2|x)}{\pi_{\text{ref}}(y_2|x)}\right)$$

$$\mathcal{L}_{\text{oss}}(\theta) = -\mathbb{E}_{x, y_1, y_2} \log \sigma\left(\beta \log \frac{\pi(y_1|x; \theta)}{\pi_{\text{ref}}(y_1|x)} - \beta \log \frac{\pi(y_2|x; \theta)}{\pi_{\text{ref}}(y_2|x)}\right) \rightarrow \min_{\theta}$$